

# Linux Course Structure

22/11/2025

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## Chapter 1: Introduction, Installation, and Basics

### Session 1: Intro

- Linux history and Linux distros
- Unix philosophy
- Kernel vs OS
- Free Software Philosophy
- Why programmers love Linux
- Pros and cons

### Session 2: Installation

Before the session : Install a VirtualBox or prepare a bootable USB.

- Choosing your distro
- Install your Linux system
- Dual booting with Windows
- VirtualBox VM setup
- Post-install (drivers – firewall – backup and restore system)

### Session 3: Basics

- System information
- Shell variables
- Commands: uname, inxi, ls, cd, cat, mkdir, cp, mv, rm, echo, sleep, pwd
- Paths: relative and absolute

- Package managers: apt, dnf, yum
- Editors: nano, gedit
- Help commands: man, info, tldr, -help, apropos
- Archiving and compression: tar, zip, 7z

## Session 4: Text Manipulation

- grep
- sed
- awk
- find
- Pipes and redirections (>, », |, <)

## Session 5: User Permissions and File System

- File system hierarchy (/ , /home , /etc , /var , /usr)
- "Everything is a file"
- Dot files
- File ownership and permissions (rwx)
- Root permissions: sudo, su, editing /etc/sudoers
- Hard links and soft links
- Common beginner mistakes

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## Chapter 2: Software and Development and Cool Stuff

Note: These are suggested topics and can be adapted.

## Session 6: Software and Development

- Compile suckless tools (st, dmenu)
- Run a simple server and share files
- PHP / XAMPP
- Node.js and basic React app
- Java basics
- Browsers, VS Code, Arduino IDE
- Compilers and debugging tools
- Makefiles and AVR toolchain
- Install Python (venv, pip)
- Use .AppImage and create a desktop shortcut
- Update system and change repositories
- Add Kali repos (with discussion)
- Install Docker and run containers

## Session 7: Cool Stuff

- Research skills and problem solving
- Learn ffmpeg
- FOSS alternatives
- Run Windows apps with Bottles and Wine/Proton/Lutris
- Install Git and clone a repo
- Learn rsync
- Using AI
- Install common GUI apps
- flatpak and snap
- Shell aliases
- Install and configure common software
- Fonts and themes

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## Chapter 3: Building and Maintaining Your System

### Session 8: Process Monitoring and Management

Check That `htop` package is installed on your system

- `htop`
- `kill` and `pkill`
- `ps` with `grep`
- Run in background (`&`)
- Command chaining (`&&`, `||`, `;`)

### Session 9: Scheduling with `systemd`

- `systemd` services
- `systemd` timers
- Basic unit files

### Session 10: Managing Disks, RAM Usage, and Swap

Check That `zram-tools` package is installed on your system

- `lsblk`
- `blkid`
- `mount` and `umount`
- Editing `/etc/fstab`
- Monitoring RAM (free)
- Adding swap or using `zram`

## Session 11: Shell Scripting and Networking Tools

Check That curl, aria2, ssh packages are installed on your system

- Bash scripting basics
- Networking tools: ping, ifconfig/ip, curl, wget, aria2, ssh, scp
- Ricing Linux with tiling window managers (optional)

### Homework for Session 11

- Write a script (bash + curl) that shows currency rates, prayer times, and weather forecast
- Install Termux (Android) or iSH (iOS) and SSH into your PC

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## Final Project

At home:

- Install Linux
- Make your own customized (riced) desktop
- Desktop environments and window managers allowed
- Share results with the group